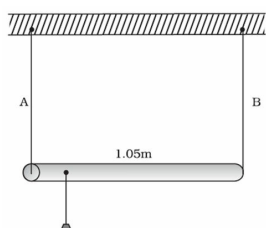


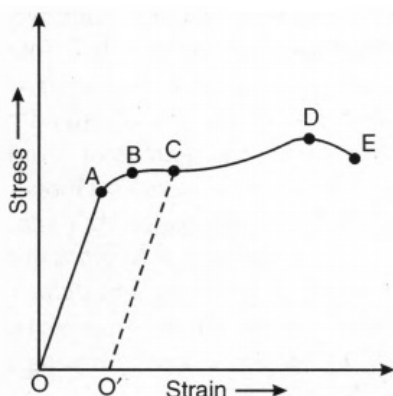
CBSE Test Paper 01
Chapter 9 Mechanical Properties of Solids

1. A solid brass sphere is initially surrounded by air, and the air pressure exerted on it is $1.0 \times 10^5 \text{ N/m}^2$ (normal atmospheric pressure). The sphere is lowered into the ocean to a depth at which the pressure is $2.0 \times 10^7 \text{ N/m}^2$. The volume of the sphere in air is 0.50 m^3 . By how much does this volume change once the sphere is submerged? modulus of brass as 61 GPa **1**
- a. $-1.7 \times 10^{-4} \text{ m}^3$
b. $-1.4 \times 10^{-4} \text{ m}^3$
c. $-1.5 \times 10^{-4} \text{ m}^3$
d. $-1.6 \times 10^{-4} \text{ m}^3$
2. For a rope of yield strength S_y loaded in tension with weight Mg the minimum area A of the rope should be **1**
- a. $A \geq Mg S_y$
b. $A \geq Mg / S_y$
c. $A \geq Mg / 3 S_y$
d. $A \geq Mg / 2 S_y$
3. A rod of length 1.05 m having negligible mass is supported at its ends by two wires of steel (wire A) and aluminum (wire B) of equal lengths as shown in Figure. The cross-sectional areas of wires A and B are 1.0 mm^2 and 2.0 mm^2 , respectively. At what point along the rod should a mass m be suspended in order to produce equal strains in both steel and aluminum wires. Take Young's modulus of steel as 200 GPa , for aluminum 70 GPa **1**



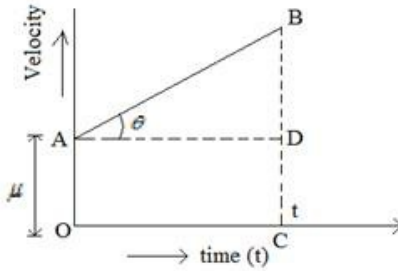
- a. 0.42 m from steel wire
b. 0.40 m from steel wire
c. 0.43 m from steel wire
d. 0.44 m from steel wire
-

4. Material is said to be brittle if **1**
 - a. material cross section is significantly reduced at failure
 - b. fracture occurs soon after the elastic limit is passed
 - c. a large amount of plastic deformation takes place between the elastic limit and the fracture point
 - d. material elongates a lot before finally breaking
5. A piece of copper having a rectangular cross-section of $15.2 \text{ mm} \times 19.1 \text{ mm}$ is pulled in tension with $44,500 \text{ N}$ force, producing only elastic deformation. Calculate the resulting strain? Take Young's modulus of copper as $11 \times 10^{10} \text{ Pa}$ **1**
 - a. 0.06×10^{-2}
 - b. 0.11×10^{-2}
 - c. 0.04×10^{-2}
 - d. 0.14×10^{-2}
6. Define Poisson's ratio? What is its unit? **1**
7. What are ductile and brittle materials? **1**
8. What is the Young's modulus for a perfect rigid body? **1**
9. Why do we prefer a spring made of steel and not of copper? **2**
10. When a load of a wire is increased from 3 kg wt to 5 kg wt , the length of that wire changes from 0.61 mm to 1.02 mm . calculate the change in the elastic potential energy of the wire. **2**
11. A wire elongates by 1 mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each is hung at the two ends, then what will be the elongation of the wire in mm ? **2**
12. The stress-strain graph for a metal wire is given in the figure. Up to the point B, the wire returns to its original state O along the curve BAO, when it is gradually unloaded. Point E corresponds to the fracture point of the wire. **3**



- Up to which point of curve, is Hooke's law obeyed? This point is also called 'Proportionality limit'.
- Which point on the curve corresponds to elastic limit and yield point of the wire?
- Indicate the elastic and plastic regions of the stress-strain curve.
- What change happens when the wire is loaded up to a stress corresponding to point C on curve, and then unloaded gradually?

13. Establish $s = ut + \frac{1}{2}at^2$ from velocity-time graph for a uniform accelerated motion? **3**



- A man carrying mass $M = 125 \text{ kg}$ makes a flying tackle at $V_j = 4 \text{ m/s}$ on a stationary quarterback of mass $m = 85 \text{ kg}$ and his helmet makes solid contact with quarterback's femur. **3**
 - What is the final speed of two athletes immediately after contact and also determine the average force exerted on the quarterback's femur, when the last collision occur at 0.100 s ?
 - If the area of cross-section of quarterback's femur is $5 \times 10^4 \text{ m}^2$, then estimate the shear stress exerted on the femur in the collision.
- Two wires of diameter 0.25 cm , one made of steel and the other made of brass are loaded as shown in Figure. The unloaded length of steel wire is 1.5 m and that of brass wire is 1.0 m . Compute the elongations of the steel and the brass wires. **5**

